

- **Information architecture**, which defines the flow of data and its internals (e.g., data modelling, data sensitivity and security).
- **Application architecture**, which defines the application landscapes, and the interfacing definitions and design patterns to be used for building the application.
- **Technological architecture**, which defines the whitelisted and blacklisted frameworks, and tools or technologies to be used.

Data architecture vs information architecture

DAMA International is a global non-profit data management community that defines various standards on information and data management. It defines data “as a collection of representation of facts as text (news), numbers (figures), graphics (datasheet), images, sound or video.” Information, according to DAMA, is: “Data in context and background. Without a context, data is absolutely meaningless or clueless to interpret; we create meaningful information by interpreting the context around data collected from different sources. The resulting information helps one to guide in taking a decision.”

Hence, data architecture (DA) is an integral component of information architecture (IA), and is concerned with the design and intent to use data in a given structured format like databases and file systems. A data architect models the data in stages (conceptual, logical and physical) and must relate this data to each process that uses it.

Information architecture (IA) is the starting point for data modelling, design, and development to support business process needs. Any new models or changes to existing models must be consistent with the definitions of existing information objects.

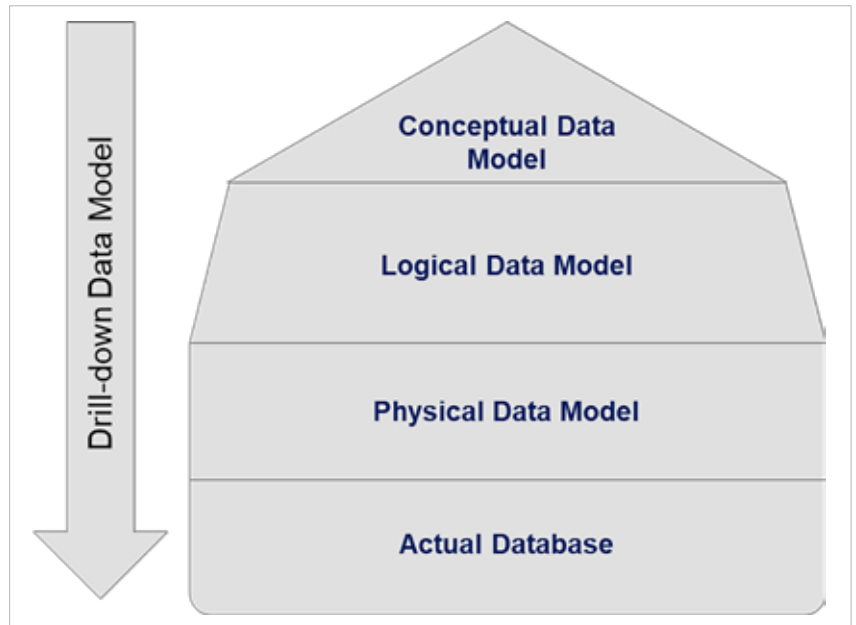


Figure 2: Stages in data modelling

Data modelling as part of EA development

In traditional architecture development, data modelling is the simple task of deriving data elements from requirements, depicting the relation between the entities through entity-relation (ER) diagrams, and defining the parameters (data types, constraints, validations) around the data elements.

In data driven architecture, where data intensity is high (such as Big Data architecture), data modelling is a crucial step (time consuming as well) and is innovatively divided into three stages.

- **Conceptual data modelling (CDM):** This is based on the business requirement to define ‘what’ is handled in the data flow. Usually, CDM is decided by business stakeholders (e.g., consultants, business owners, application analysts) and data architects.
- **Logical data modelling (LDM):** This is the next step after CDM. It works on the logical relations between the entities and details their data types. It defines ‘how’ data is handled in the data flow, which is

decided by business consultants and data architects.

- **Physical data modelling (PDM):** The actual blueprint is built here based on LDM, which gets translated into data scripts for execution in a live environment. This crucial stage, where the data structure, transaction handling mechanism, tuning and optimisation of the data model is implemented, is typically handled by a database analyst or data engineer.

Data driven modelling helps to build a strong foundation for architecture design. It aids in designing ‘fail fast’ architecture — building the right architecture in the early stages and reducing failure at later stages.

Popular tools for EA development

- **Enterprise Architect** Sparx Systems’ Enterprise Architect is a comprehensive, multi-functional modelling tool that has been a staple in the enterprise architecture space for many years. Known for its robust support for a wide variety of modelling languages